BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 9231

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** 9231

Note: Attempt all questions of Section-A by filling the corresponding bubble on the MCQs RESPONSE SHEET. It is mandatory to return the attempted MCQs sheet to the Superintendent within given time.

Q1. The value of gravitational field strength is maximum at:

(A) Equator
(B) Pole
(C) Surface of Earth
(D) Center of Earth

Q2. A truck covers a distance of 20 km with a constant force of 400 N, its work done is:

(A) 20 kJ
(B) 2000 kJ
(C) 4000 kJ
(D) 8000 kJ

Q3. Capacity of a body to do work is:

(A) Force
(B) Power
(C) Energy
(D) Momentum

Q4. Fluids exert pressure in:

(A) Upward direction only
(B) Downward direction only
(C) Two directions
(D) All directions

Q5. The quantity of heat required to raise the temperature of a substance by 1°C or 1 K is known as:

(A) Specific Heat
(B) Latent Heat
(C) Heat Capacity
(D) Internal Energy

Q6. Conduction occurs best in:

(A) Solids
(B) Liquids
(C) Gases
(D) Plasma

Q7. Which of the following is base quantity?

(A) Force
(B) Electric Current
(C) Density
(D) Energy

Q8. Displacement becomes equal to distance, when the path followed by moving body is:

(A) Circular
(B) Zigzag
(C) Straight
(D) Curved

Q9. An agent which moves or tends to move a body is called:

(A) Momentum
(B) Inertia
(C) Force
(D) Impulse

Q10. The magnitude of centripetal force acting on a stone of mass 1 kg, moving in a circle of radius 3 m with speed 3 m/s is:

(A) 3N
(B) 4N
(C) 5N
(D) 6N

Q11. Ali is pushing a block with a force of 40 N at 60° with floor, horizontal component of force is:

(A) 20 N
(B) 30 N
(C) 35 N
(D) 40 N

Q6. Turning effect on a door can be increased by increasing:

(A) Mass
(B) Moment Arm
(C) Momentum
(D) Inertia

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 32 | **Paper Code:** 9231

Q1. Attempt any eight of the following short questions. Each question carries 4 marks

(i) Define the following branches of Physics: (i) Atomic & Nuclear Physics (ii) Thermodynamics (iii) Quantum Physics (iv) Mechanics

(ii) Differentiate between vectors and scalars by giving one example of each?

(iii) If a force 120 N act on a body its momentum changes by 60kg m/s. What is the time for which the force acts on the body?

(iv) What is uniform circular motion? Give example.

(v) Write four steps of head to tail rule for finding resultant of forces?

(vi) Define equilibrium. What is the difference between static and dynamic equilibrium?

(vii) What is meant by "Bio-Energy". Explain how solar energy is converted into "Bio-Energy?"

(viii) Define Power. A crane is capable of doing 2.50×10^5 J work in 10 s. What is the power of crane?

(ix) Define elasticity. State Hook's law and give its mathematical form.

(x) Define evaporation. How nature of liquid effects the rate of evaporation?

(xi) Define convection of heat. Explain the significance of convection currents in heating water.

SECTION-C

Maximum Marks: 21

Note: Attempt any three of the following questions. Each question carries 7 marks

Q2.

(i) Use graphical method to derive 3rd equation of motion.

(ii) An aircraft takes off at 82 m/s after accelerating from rest at 2.80 m/s^2. What is the minimum runway length required?

Q3.

(i) Calculate the mass of earth.

(ii) The mass of Mars is 6.4×10^{23} kg and having radius of 3.4×10^6 m. Calculate the gravitational field strength(g) on Mars surface.

Q4.

(i) State and explain Archimedes Principle.

(ii) A geologist find that a rock whose weight is 90.9 N has an apparent weight of 60.6 N when submerged in water what is the density of rock?

Q5.

(i) What is meant by linear thermal expansion? Derive expression for coefficient of linear thermal expansion.

(ii) A railway line made of iron is 1200 km long laid at 25°C. By how much will it contract in winter when the temperature falls to 15°C?